

Linearly Independent & Dependent Vectors : -

Example : $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Then $3v_1 + 2v_2 = ?$

$$\rightarrow 3v_1 + 2v_2 = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Example : $v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Then $5v_1 - 2v_2 = ?$

$$\rightarrow 5v_1 - 2v_2 = 5 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ -2 \end{bmatrix}$$

Example : $x_1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 9 \end{bmatrix}$ ← [Hint: System of linear equation]

Find x_1, x_2 and x_3

→ Let's try $x_1 = 1, x_2 = 4, x_3 = -2$

$$1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -4 \\ -6 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 9 \end{bmatrix}$$

* Linear Combination of Vectors

→ A linear combination of vectors is just a weighted sum of vectors

* Formal definition:

↳ Given vectors $v_1, v_2, \dots, v_k$

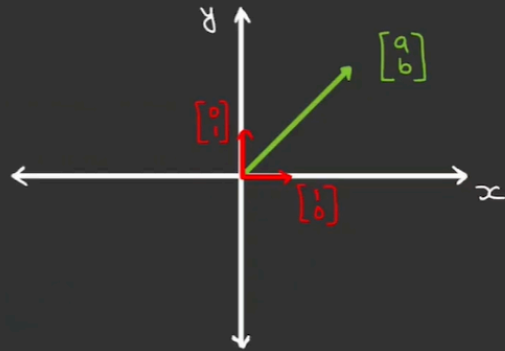
Any vector of the form $u = a_1v_1 + a_2v_2 + \dots + a_kv_k$

$x_3 = LC(\{v_1, v_2\})$
 $x_3 = a_1v_1 + a_2v_2$
 $a_1, a_2 \leftarrow \text{coeffs}$

is called a linear combination of those vectors where $a_1, a_2, a_3, \dots, a_k$ are scalars.

* Example:

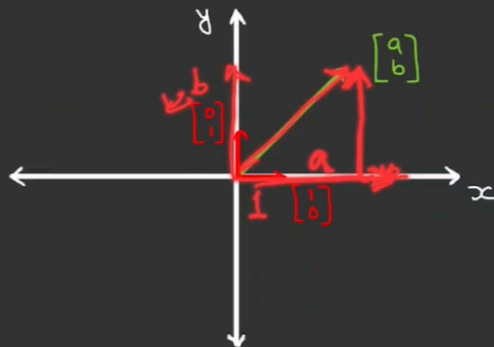
$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



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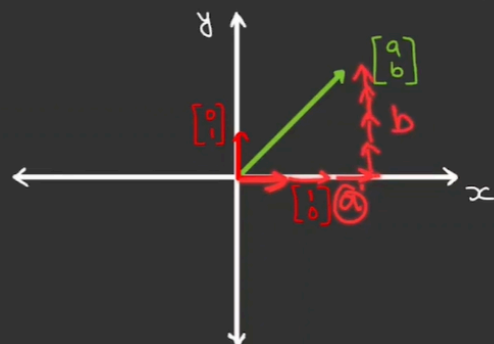
$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



is called a linear combination of those vectors where $a_1, a_2, a_3, \dots, a_k$ are scalars.

* Example:

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



* Geometric Intuition:

→ You scale vectors (stretch/shrink). Then add them.

→ In machine Learning, almost everything is a Linear combination.

$$\hat{y} = w_1 x_1 + w_2 x_2 + \dots + w_n x_n \leftarrow \text{Housing Price example}$$

↳ here prediction is a linear combination of Input features.

* Linearly dependent vectors

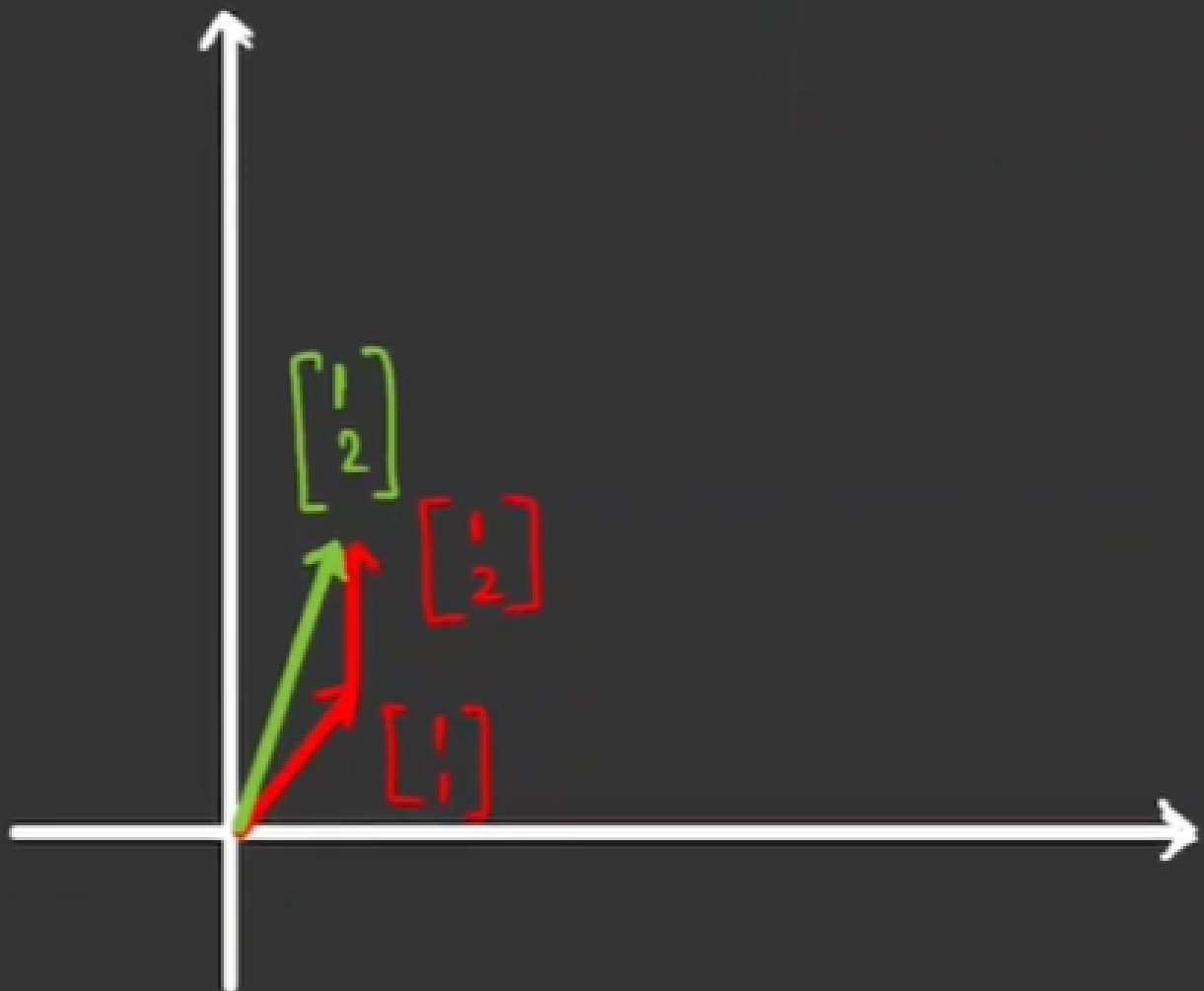
→ Linear dependence is defined for set of vectors.

$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ is an example of set of vectors.

→ A set of vectors is Linearly dependent if atleast one vector can be written as a combination of others.

Example: $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ are Linearly dependent?

→ yes ($w = u + v$)



@Important Point : When features in a dataset are linearly dependent or strongly correlated, they introduce redundancy. Eliminating or transforming these redundant features into linearly independent ones is key to building clean, efficient models while reducing computational overhead, training time, and complexity.

Example: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 4 \end{bmatrix} \right\}$ are Linearly dependent?

→ yes ($u + v = w$)

Example: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 12 \\ 3 \end{bmatrix} \right\}$ are Linearly dependent?

→ yes ($w = 0u + 3v$)

Example: $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$ are Linearly dependent?

Example: $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$ are Linearly dependent?

→ yes ($u = 0v + 0w$)

Example: $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \begin{bmatrix} d \\ e \\ f \end{bmatrix} \right\}$ are Linearly dependent?

→ yes ($u = 0v + 0w$)

A set containing zero vector is always linearly dependent.

→ If a set of vectors are Linearly dependent $\Rightarrow$ There is atleast one vector that can be represented as a linear combination of other vectors.

→ Mathematically, A set of vectors are Linearly dependent if

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_k v_k = 0$$

has non-zero solution (some $c_i \neq 0$)

Intuition:

Condⁿ for LD: $v_1 = a_2 v_2 + a_3 v_3 + \dots + a_k v_k$

$$\textcircled{-1}_{a_1} v_1 + a_2 v_2 + a_3 v_3 + \dots + a_k v_k = 0 \Rightarrow a_1 = -1$$

Example: Given, $v_1 = c_2 v_2 + c_3 v_3 + c_4 v_4 + \dots + c_n v_n$

Can we represent v_3 as a linear combination of other vectors?

$$\rightarrow c_3 v_3 = 1 \cdot v_1 - c_2 v_2 - c_4 v_4 - \dots - c_n v_n$$

$$v_3 = \frac{1}{c_3} v_1 - \frac{c_2}{c_3} v_2 - \frac{c_4}{c_3} v_4 - \dots - \frac{c_n}{c_3} v_n$$

If $c_3 \neq 0$, we can represent v_3 as a linear combination of other vectors.

Linearly Independent Vectors :

→ A set of vectors are Linearly Independent iff they are not Linearly dependent.

→ A set of vectors is linearly Independent if no vectors can be written as a combination of the others.

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→ Mathematically $c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_k v_k = 0$

has only zero-solution (all $c_i = 0$)

→ Intuition:

① Each vector add new information

② No redundant information

Example: If a set is Linearly dependent, then the superset is Linearly dependent. (True)

→ Let $\{v_1, v_2, v_3\}$ be a LD set and $\{v_1, v_2, v_3, v_4\}$ be superset

$$\hookrightarrow v_1 = a_2 v_2 + a_3 v_3$$

$$\hookrightarrow v_1 = a_2 v_2 + a_3 v_3 + 0 v_4$$

↳ so $\{v_1, v_2, v_3, v_4\}$ is LD set

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Example: If a set is Linearly dependent, then the subset is Linearly dependent. (False) ✓

→ let $\{v_1, v_2, v_3\}$ be a LD set such that $v_1 = a_2 v_2 + a_3 v_3$

↳ and let $\{v_2, v_3\}$ be a subset

↳ Can't say

$$\hookrightarrow \left\{ \underset{v_1}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}, \underset{v_3}{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \right\} \text{ be a set such that } v_1 = v_2 + v_3$$

↳ and let $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ be a subset

↳ It is linearly independent

$\hookrightarrow \left\{ \underset{v_1}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}, \underset{v_3}{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \right\}$ be a set such that $v_1 = v_2 + v_3$

↳ and let $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ be a subset

↳ It is linearly independent